

RGB LED Control Using Push Buttons on Tiva C Series

Name: Uma Bharathi D

SR Number: 25964

Programme: M. Tech ESE

Overview

This project is implemented on the Tiva C Series TM4C123GH6PM microcontroller. It demonstrates RGB LED control using two onboard push buttons. One switch is used to change the LED color, while the other switch is used to vary the LED blink rate. The program uses GPIO configuration, software delays, and debounce logic for reliable operation.

Features

- RGB LED control using GPIO Port F
- Color cycling using SW1 (Red → Green → Blue)
- Blink rate adjustment using SW2
- Software-based debounce for push buttons
- Software-controlled LED blinking

Hardware Requirements

1. Tiva C Series TM4C123GH6PM LaunchPad
2. Onboard RGB LED (PF1, PF2, PF3)
3. Onboard push buttons SW1 (PF4) and SW2 (PF0)
4. USB cable for power and programming

GPIO Pin Configuration

Port	Pin(s)	Function
Port F	PF1, PF2, PF3	RGB LED output
Port F	PF4	SW1 – Color change (active low)
Port F	PF0	SW2 – Blink delay change (active low, unlocked)

Working Principle

The GPIO Port F is configured for both input and output operations. The RGB LED pins are configured as outputs, while the push button pins are configured as inputs with internal pull-up resistors. When SW1 is pressed, the LED color cycles through predefined colors.

When SW2 is pressed, the delay count controlling the blink rate is updated. A software counter and delay function are used to generate the LED blinking behavior.

Conclusion

This project demonstrates effective use of GPIO, software delays, and switch debouncing on the TM4C123GH6PM microcontroller. It provides a clear understanding of embedded C programming for real-time LED control using push buttons.